Dual-Directional Buffered PLUS linked to the S&P 500 Index

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Executive summary: We value Morgan Stanley Finance LLC's Dual-Directional Buffered PLUS linked to the S&P 500 Index (with issue price \$1,000; valuation time 30-Dec-2025 16:00; fixing date 30-Dec-2027; payment on 04-Jan-2028). Using USD SOFR OIS discounting, Bloomberg OVME/BVOL implied forward and implied volatilities as of 30-Dec-2025 16:00 (underlying set to "Last"), and a Black–Scholes static-replication framework with cashflow-date alignment, our base-case estimated value is **\$977.88** per note. This compares with the issuer's stated estimated value of \$973.20 (difference +4.68). The dominant sensitivity remains left-tail implied volatility near the 90% buffer strike.

1. Product and payoff description

The note references the S&P 500 Index with initial level S_0 = 6,896.24. Payoff is determined on the fixing date (30-Dec-2027) and paid on the payment date (04-Jan-2028). Key terms: buffer = 10% ($K_B = 0.9 \cdot S_0$), leveraged upside = 150%, minimum payment = \$100, and maximum payment = \$1,176.70 (cap). Let $K_C = (1 + 0.1178) \cdot S_0 = 7{,}708.62$.

- If $S_T \geq K_C$: 1,176.70
- If $S_0 \leq S_T < K_C$: $1{,}000 + 1{,}500 \cdot (S_T/S_0 - 1)$
- If $K_B \leq S_T < S_0$: $2{,}000 - 1{,}000 \cdot (S_T/S_0)$
- If $S_T < K_B$: $1{,}000 \cdot (S_T/S_0) + 100$

A \$100 discontinuity at K_B makes the payoff digital-type at the buffer boundary.

2. Data and market inputs

Item	Value
Initial index level S_0	6,896.24
Fixing date t_2	30-Dec-2027
Payment date t_3	04-Jan-2028
OIS DF(t_0,t_2) (fixing)	0.936643
OIS DF(t_0,t_3) (payment)	0.936214
Cashflow-date adjustment Adj = DF(t_0,t_3)/DF(t_0,t_2)	0.999542
Forward to fixing F	7,186.50
Implied dividend yield q (continuous)	1.234%
Implied vol (K/S_0=0.895)	21.668%
Implied vol (K/S_0=0.900) [buffer]	21.547%
Implied vol (K/S_0=0.905)	21.417%
ATM implied vol (K/S_0=1.000)	18.942%
Implied vol at cap strike (K/S_0=1.1178)	16.074%

Discount factors are taken from the USD SOFR OIS curve anchored at t_0=30-Dec-2025 (16:00). OVME option PVs are quoted assuming cash settlement on the fixing/expiry date t_2, while the note pays on t_3; we align cashflow dates by scaling option PVs by $Adj = DF(t_0, t_3)/DF(t_0, t_2)$. Given the ~5-day lag, Adj is very close to 1 but removes a cashflow-date mismatch.

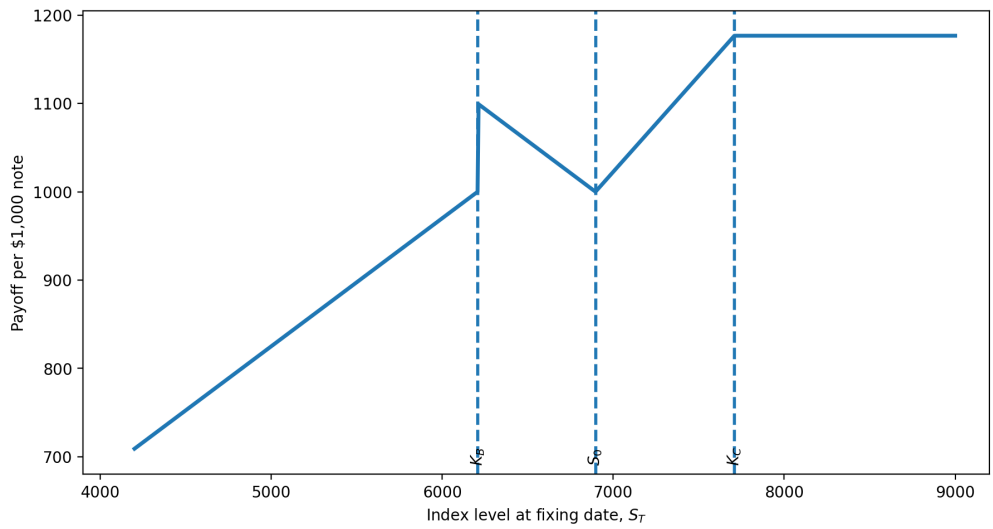

Figure 1: Note payoff as a function of the index level at the fixing date, S_T.

3. Static replication

The payoff is piecewise linear with kinks at K_B, S_0, and K_C. Matching slope changes across regions yields the exact decomposition into a cash bond, a prepaid forward (linear exposure), plain-vanilla calls, and a digital at K_B:

$$\text{Payoff}(S_T) = 100 + \frac{1000}{S_0} S_T - \frac{2000}{S_0}(S_T - K_B)^+ + 100\,\mathbf{1}_{\{S_T \geq K_B\}} + \frac{2500}{S_0}(S_T - S_0)^+ - \frac{1500}{S_0}(S_T - K_C)^+.$$

The linear term $(1000/S_0)S_T$ is implemented as a prepaid forward (PV of receiving S_T at t_3). A tradable representation uses put-call parity at strike K: $\Pi_{\text{pfwd}}(K) = C(K) - P(K) + K \cdot DF(t_0, t_3)$.

Component	Description	Position (units)
Cash	Pay \$100 at payment date t_3	100.0
Prepaid forward	Long PV of receiving S_T at t_3 (linear exposure)	0.145007
Call	Short $(2000/S_0)$ calls, $K = K_B = 6{,}206.616$	-0.290013
Digital	Long \$100 cash-or-nothing digital at K_B (pays 1 if $S_T \geq K_B$)	100.0
Call	Long $(2500/S_0)$ calls, $K = S_0 = 6{,}896.24$	0.362516
Call	Short $(1500/S_0)$ calls, $K = K_C = 7{,}708.62$	-0.217510

4. Valuation and results

We value each replication term under Black-Scholes using the OIS discount factor to the payment date $DF(t_0, t_3)$ and the equity forward F for the fixing date t_2. Bloomberg OVME "model prices" correspond to Black's formula with strike-specific BVOL implied vols $\sigma(K)$. Because OVME option PVs are quoted assuming cash settlement on t_2, we convert them to payment-date PVs via $PV_{\text{pay}} = PV_{\text{fix}} \cdot Adj$, where $Adj = DF(t_0, t_3)/DF(t_0, t_2)$.

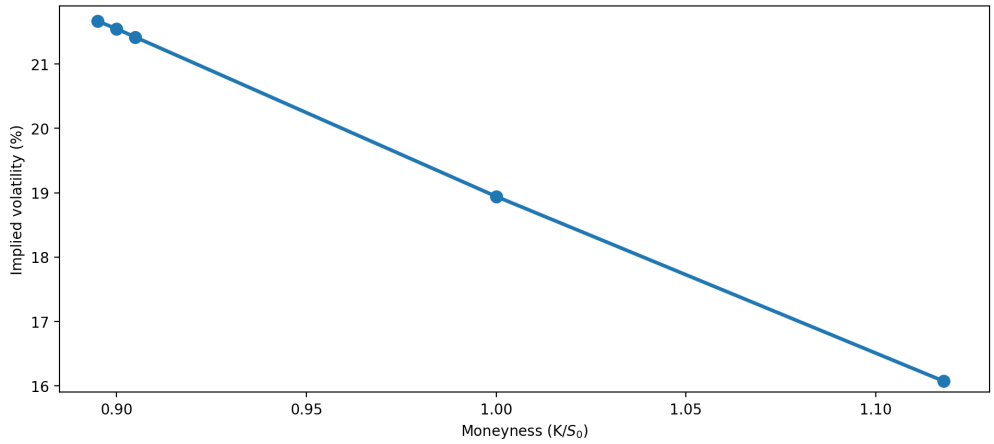

Figure 2: OVME/BVOL implied vol smile (24M) used for valuation.

Digital leg (cash-or-nothing, amount $A = \$100$ at t_3). Under strike-dependent implied vol $\sigma(K)$, the digital price equals the Black-Scholes baseline plus a first-order skew correction (from $-dC/dK$):

$$D = A \cdot DF \cdot N(d_2) + A \cdot (-\text{Vega}) \cdot \left(\frac{\partial \sigma}{\partial K}\right).$$

Using $\sigma(K_B) = 21.547\%$, $T \approx 2y$, $F = 7{,}186.50$, we obtain $N(d_2) = 0.6288$ and Vega $= 3{,}105.94$. The local skew is estimated by

$$\frac{\partial \sigma}{\partial K} \approx \frac{\sigma(0.905S_0) - \sigma(0.895S_0)}{0.01 \cdot S_0} = -0.00363\% \text{ per index point.}$$

This gives digital PV = 58.87 (baseline) + 11.28 (skew) = **70.15**.

Term	DF / Adj	Market/Model input	PV contribution ($)
\$100 cash	DF=0.936214	–	93.62
$(1000/S_0) \cdot S_T$ (prepaid forward)	DF=0.936214	F=7,186.50	975.62
$-(2000/S_0) \cdot \text{Call}(K_B)$	Adj=0.999542	OVME C_{fix}=1,303.81	-377.95
$+100 \cdot \text{Digital}(K_B)$	DF=0.936214	BS digital + vega×skew	70.15
$+(2500/S_0) \cdot \text{Call}(S_0)$	Adj=0.999542	OVME C_{fix}=847.06	306.93
$-(1500/S_0) \cdot \text{Call}(K_C)$	Adj=0.999542	OVME C_{fix}=416.23	-90.49
		Total model value	**977.88**

Model value: \$977.88 — Issuer estimated value: \$973.20 (difference +4.68).

5. Sensitivity and accuracy discussion

Key sources of uncertainty are (i) left-tail implied vol near the 90% buffer strike, (ii) the local implied-vol skew used in the digital correction, and (iii) the implied dividend/carry embedded in the forward.

Skew estimation sensitivity. We estimate the local skew slope $\partial\sigma/\partial K$ around K_B using backward/central/forward differences of the 24M vol quotes, and update only the digital skew adjustment (baseline held fixed), holding all other inputs and replication legs fixed.

Skew-slope estimator (24M vols)	$\partial\sigma/\partial K$ **(% per index pt)**	**Digital PV ($)**	**Note value ($)**
Backward diff ($0.895S_0$, $0.900S_0$)	-0.00352%	69.79	977.52
Central diff ($0.895S_0$, $0.905S_0$)	-0.00363%	70.15	977.88
Forward diff ($0.900S_0$, $0.905S_0$)	-0.00375%	70.52	978.25

Left-tail volatility sensitivity. We shift the 24M left-tail vol quotes at $K/S_0 = \{0.895, 0.900, 0.905\}$ in parallel and reprice the K_B call and the digital using the Black (forward-based) formula, holding all other vols and replication legs fixed.

σ_B **(left-tail) %**	**Note value ($)**
19.0	1,003.04
20.0	993.20
21.547	977.88
22.0	973.38
23.0	963.40

Dividend / forward sensitivity. We implement changes in the dividend yield q by updating the implied forward F (holding the financing rate and the OIS discount curve fixed), while keeping all implied vols and other replication legs unchanged.

Dividend yield q **(%)**	**Forward** F	**Note value ($)**
0.734	7,258.75	983.03
0.984	7,222.55	980.47
1.234	7,186.52	977.89
1.484	7,150.68	975.28
1.734	7,115.02	972.65

Conclusion: Our estimated economic value is **$977.88** per $1,000 note using OVME/BVOL matrix inputs at 30-Dec-2025 16:00, with cashflow-date alignment and a skew-corrected digital. A plausible sensitivity-implied submission range is about $963—$1,003, driven primarily by uncertainty in left-tail implied volatility and secondarily by carry/dividend assumptions.

References: Morgan Stanley Finance LLC Form 424B2; Bloomberg Terminal (OVME, BVOL matrix, ICVS) as of 30-Dec-2025 16:00.